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AI-Based Predictive Maintenance for Induction Motors using a Hybrid CNN-LSTM Architecture with Attention Mechanism

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***Abstract*—In Industrial rotating machinery, predictive maintenance plays a significant role in reducing downtime and the cost of operation. The proposed paper suggests a deep-learning-based fault detection of induction motors, utilizing the hybrid architecture, which combines a Convolutional Neural Network (CNN) with a Bidirectional Long Short-Term Memory (BiLSTM) network, enhanced with an attention mechanism. The model learns the multi-scale spatial features by using a multi-kernel CNN branch and the temporal relationship at the same time by using a stacked BiLSTM network. An attention layer provides selective emphasis on the most informative temporal instants and increases the classification accuracy. The suggested framework is tested on the Fault Induction Motor Dataset, where the test accuracy is 98.60%, and the precision, recall, and F1-score are more than 0.986. Moreover, a real-time monitoring dashboard is adopted to visualize system status and predictions. The empirical results support the usefulness of the hybrid method in the correct classification of six different motor conditions.**

***Index Terms*—Predictive Maintenance, Deep Learning, CNN-LSTM, Attention Mechanism, Induction Motor, Fault Diagnosis**

I. Introduction

Induction motors are a cornerstone of the modern industrial system, providing actuation for equipment in the manufacturing, transportation, and energy generation sectors. The failure of these vital assets that are not predicted results in expensive production disruptions, safety threats, and significant financial losses, and downtime costs range from a few thousand to several million dollars per hour, depending on the industry involved.

Traditional paradigms in maintenance have advanced in the progression from reactive (run-to-failure) to preventive (time-based) maintenance. However, the two approaches are usually too costly or ineffective. Predictive maintenance (PdM) offers a more data-driven alternative, which uses sensor streams and machine-learning methods to predict equipment degradation and plan interventions ahead of time based on real operating conditions instead of scheduled maintenance [1].

The growth of Industrial Internet of Things (IIoT) systems has enabled the acquisition of high-frequency vibration, current, and temperature measurements of the motors in a continuous manner. These signals are analyzed to identify the early faults, and this is a challenging pattern recognition problem, due to the non-stationary nature of the data and variability of the operating environment. The classical methods relied on engineered features and classic classifiers like Support Vector Machines (SVM) [2]. Effective, but these methods require extensive domain knowledge, and they might not be able to generalize under different conditions.

Deep learning has become an effective technique of automatic feature extraction and sequential modelling. Convolutional Neural Networks (CNNs) are skilled at the detection of local patterns, which makes them the right choice to detect fault-related patterns [3]. Recurrent Neural Networks (RNNs) and especially Long Short-Term Memory (LSTM) networks can be especially efficient at learning long-term temporal dependencies [4]. Hybrid CNN-LSTM systems take advantage of these complementary capabilities to classify time-series [5].

This paper presents a new hybrid deep-learning model that is specific to the diagnosis of faults in induction-motors. The main contributions are:

1) A multi-scale feature-extraction module that consists of three parallel CNN branches with the kernel sizes of 3, 5, and 7. To capture fault signatures at different temporal resolutions, we employed a range of kernel sizes—smaller kernels to resolve high-frequency transients and larger kernels to capture sustained fault patterns.
2) A BiLSTM stacked network that would enable the network to capture both past and future time-step context.
3) A dynamically changing mechanism of assigning weights to the importance of individual time steps.
4) Our complete feature-engineering pipeline extracts more than thirty features spanning statistical, frequency-domain, and wavelet-based categories. We further examine how these handcrafted features work alongside deep learning features to improve overall performance.
5) A monitoring dashboard to be used in practice.

The rest of this paper is structured in the following way: Section II is the review of related literature. Section III elaborates on the proposed methodology. Section IV provides the implementation and experimental configuration. Section V gives the empirical results and provides a discussion. Section VI provides closing comments.

II. Related Work

A. Traditional Machine Learning Approaches

Initial studies were aimed at deriving features out of sensor signals and then classifying them with machine learning algorithms. Some of the feature extraction methods were statistical parameters (mean, RMS, kurtosis, skewness), statistical frequency domain analysis based on Fast fourier Transform (FFT) and time-frequency analysis using wavelet transforms [12].

Widodo and Yang [2] presented an overall overview of machine learning approaches to condition monitoring and discussed feature extraction, dimensionality reduction, and classifiers such as SVMs and neural networks. They emphasized the significance of selecting features and addressing issues with generalizing across operating conditions.

The proposal that the signature of motor current analysis (MCSA) can be used to detect broken rotor bars and bearing faults was investigated by Schoen et al. [13], who showed that the frequencies of faults are manifested as sidebands around the base supply frequency.

B. Deep Learning for Fault Diagnosis

Deep learning has facilitated end-to-end learning on raw data, which does not require manual feature engineering. Inoue et al. [3] used a deep CNN to process raw vibration data to classify bearing faults and obtained more than 99% accuracy on the Case Western Reserve University dataset.

Zhang et al. [6] suggested a deep CNN using wide first-layer kernels to perform bearing fault diagnosis in noisy environments, which demonstrated great performance with large amounts of noise.

Reconfigurable architectures have been considered to represent time evolution of faults. Zhao et al. [7] used LSTMs to predict the remaining useful life (RUL), demonstrating that LSTMs have the ability to identify the degradation patterns of sensor data in a multivariate setting.

Graves et al. [8] proposed a Bidirectional LSTM (BiLSTM) based speech recognition system, which proved that both forward and backward processing sequences have more complete contextual information.

C. Hybrid and Attention-Based Models

Hybrid CNN-LSTM networks have the advantages of both architectures. Karim et al. [5] introduced a fully convolutional network with LSTM on time-series classification, and it delivers state-of-the-art results on benchmark datasets.

Shao et al. [9] created a CNN-LSTM motor fault diagnosis model extracting spatial information using a CNN and temporal dynamics using an LSTM, with 97.5% accuracy.

Attention mechanisms, which were first introduced in machine translation [10], have been used in time-series analysis. Attention was used by Chen et al. [11] to predict RUL of lithium-ion batteries, showing the time steps that best predicted future degradation.

Our work extends existing methods by combining multi-scale CNN feature extraction, stacked BiLSTMs, and attention mechanisms within a single architecture—offering both improved performance and interpretability. Unlike prior approaches that focus primarily on single-scale feature extraction, our method captures fault signatures across multiple temporal resolutions, producing a more robust feature representation.

D. Recent Advances (2024–2026)

Recent literature has witnessed significant advancements in induction motor fault diagnosis through hybrid deep learning architectures and innovative data-efficient approaches. Jakaria et al. [14] evaluated eight hybrid architectures including CNN-GRU and CNN-LSTM on a dataset of one million samples, achieving accuracies of 92.57% and 92.27% respectively with superior computational efficiency for real-time deployment. Addressing data scarcity, Vo et al. [15] introduced a Dual-Stream Attention-GAN framework utilizing N4SID-based residual preprocessing, achieving 99.61% accuracy on CWRU benchmark and 97.40% on industrial datasets with only 10% labeled data. Ali et al. [16] proposed the Multimodal Hypergraph Contrastive Attention Network (MM-HCAN), integrating contrastive learning with hypergraph topology for multimodal fault diagnosis, achieving up to 99.82% accuracy with strong cross-domain generalization. The integration of digital twin technology has emerged as transformative, with Phutane et al. [17] reporting up to 25% higher fault detection accuracy and 15–30% downtime reduction compared with traditional methods. Transfer learning techniques reviewed by Kumar [18] have demonstrated fault detection accuracies exceeding 99% with significantly reduced training data requirements. For synchronous and permanent magnet motors, Xu et al. [19] achieved diagnostic accuracy not less than 97% using CNN-LSTM-Attention, while Ozturk et al. [21] developed an optimized CNN-GRU framework with WOALN hyperparameter optimization achieving 100% classification accuracy. Lightweight architectures suitable for edge deployment have also gained attention, with Phutane et al. [20] implementing ShuffleNetV2 with transfer learning for broken rotor bar diagnosis at 98.856% accuracy, and Barrera-Llanga et al. [22] comparing six CNN architectures, with VGG19 achieving 99% accuracy and F1-scores ranging from 0.994 to 0.998. These recent studies collectively demonstrate the ongoing evolution toward hybrid architectures combining spatial and temporal feature extraction, attention mechanisms, multimodal fusion, and data-efficient learning strategies—aligning closely with the contributions of the proposed work.

III. Methodology

A. System Architecture Overview

The proposed system includes three stages, which are the data preprocessing and feature engineering, the hybrid CNN-

LSTM-Attention model, and a real-time monitoring dashboard.

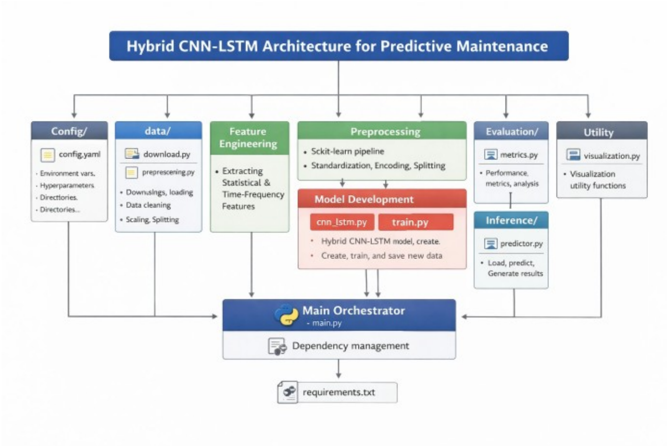

Fig. 1. System architecture of the proposed predictive maintenance framework

B. Data Preprocessing and Feature Engineering

The dataset is preprocessed with the removal of missing values by use of median imputation, encoding of categorical labels and SMOTE to overcome imbalance in classes. StandardScaler is used to make features normalized in terms of zero mean and unit variance.

Each time-series signal has 32 features extracted in a detailed feature engineering pipeline:

Time-domain features (12 features): Mean, standard deviation, RMS, peak, peak-to-peak, skewness, kurtosis, crest factor, shape factor, impulse factor, median, and signal energy.

Frequency-domain features (15 features): The FFT is computed as:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N} \tag{1}$$

Features include dominant frequency, spectral centroid, spectral spread, spectral skewness, spectral kurtosis, and energy in three frequency bands (0-50Hz, 50-200Hz, 200-500Hz).

Wavelet-based features (5 features): Wavelet decomposition is simplified to provide approximation and detail coefficients, out of which statistics and ratios of energies are derived.

C. Hybrid CNN-LSTM-Attention Model

The main innovation is the hybrid model that is a combination of three connected branches.

1) Multi-Scale CNN Branch: Three parallel 1D convolutional blocks with kernel sizes of 3, 5, and 7 capture fault signatures across varying time resolutions. The choice of these specific kernel sizes is determined by the nature of the fault signatures they target. A kernel size of 3 captures high-frequency transient faults, such as bearing impacts; kernel size 5 identifies medium-duration patterns, including stator fault harmonics; and kernel size 7 extracts sustained fault characteristics, such as broken rotor bar oscillations. This multi-scale strategy ensures thorough feature extraction across varied fault types. Each block follows the same structure:

1) Conv1D(64, kernel_size=k, activation='relu', padding='same')
2) BatchNormalization()
3) MaxPooling1D(pool_size=2)
4) Dropout(0.3)
5) Conv1D(128, kernel_size=k, activation='relu', padding='same')
6) BatchNormalization()
7) MaxPooling1D(pool_size=2)
8) Dropout(0.3)
9) Conv1D(256, kernel_size=k, activation='relu', padding='same')
10) BatchNormalization()
11) GlobalAveragePooling1D()

The convolution operation is represented as:

$$\mathbf{h}^l = \sigma\left(\mathbf{W}^l * \mathbf{h}^{l-1} + \mathbf{b}^l\right) \tag{2}$$

Batch normalization stabilizes training:

$$\hat{h} = \frac{h - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \cdot \gamma + \beta \tag{3}$$

2) Bidirectional LSTM Branch with Attention: This input is reformulated in the sequence modeling. This sequence is fed through a three-layer stacked BiLSTM network:

First BiLSTM layer: 128 units, returns sequences. Forward and backward LSTMs capture past and future contexts:

$$\overrightarrow{h}_t = \mathrm{LSTM}_{\mathrm{fwd}}(x_t, \overrightarrow{h}_{t-1}) \tag{4}$$

$$\overleftarrow{h}_t = \mathrm{LSTM}_{\mathrm{bwd}}(x_t, \overleftarrow{h}_{t+1}) \tag{5}$$

$$h_t = [\overrightarrow{h}_t; \overleftarrow{h}_t] \tag{6}$$

LSTM cell dynamics are governed by:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{7}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{8}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{9}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{10}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{11}$$

$$h_t = o_t \odot \tanh(C_t) \tag{12}$$

Attention mechanism: The attention mechanism assigns relevance weights to each time step, directing the model's focus toward diagnostically significant temporal segments. These attention scores are obtained through a dense layer and normalized using softmax:

$$e_t = \tanh(W_a h_t + b_a) \tag{13}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{14}$$

The context vector is computed as a weighted sum of hidden states:

$$c = \sum_{t=1} \alpha_t h_t \tag{15}$$

This context vector highlights time steps with higher fault-related information and suppresses irrelevant segments.

Second BiLSTM layer: 64 units.

Third BiLSTM layer: 32 units, producing the final temporal representation.

3) Feature Fusion and Classification: The final BiLSTM layer (size 64) and three CNN branches (size 256), give output vectors that are concatenated to yield a feature Vector of size 832. The classification head comprises of:

1) Dense(256, activation='relu', kernel_regularizer=l1_l2(1e-5, 1e-4))
2) BatchNormalization()
3) Dropout(0.5)
4) Dense(128, activation='relu', kernel_regularizer=l1_l2(1e-5, 1e-4))
5) BatchNormalization()
6) Dropout(0.5)
7) Dense(64, activation='relu')
8) BatchNormalization()
9) Dense(6, activation='softmax')

Softmax gives class probabilities:

$$P(y = j \mid \mathbf{x}) = \frac{e^{\mathbf{w}_j^T \mathbf{x} + b_j}}{\sum_{k=1}^{6} e^{\mathbf{w}_k^T \mathbf{x} + b_k}} \tag{16}$$

D. Training Strategy

This model is trained using Adam optimizer (learning rate 0.001, gradient clipping 1.0) and categorical cross-entropy loss:

$$\mathrm{L} = -\sum_{i=1}^{N} \sum_{j=1}^{6} y_{ij} \log(\hat{y}_{ij}) \tag{17}$$

Based on experimental validation, the number of epochs was set to 30, as convergence occurred after approximately 25 epochs and validation metrics stabilized afterwards. To prevent overfitting, early stopping with a patience of 15 epochs was applied. The training used the following callbacks: EarlyStopping (patience = 15), ReduceLROnPlateau (factor = 0.5, patience = 7), ModelCheckpoint, and CSVLogger.

Class weights address residual imbalance:

$$w_j = \frac{N}{n_{\text{classes}} \cdot N_j} \tag{18}$$

The training was conducted on an Intel Core i7-10750H with 16GB RAM and NVIDIA GPU acceleration. Total training time was approximately 45 minutes for 30 epochs.

IV. Implementation and Experiments

A. Dataset

The system is tested on the Fault Induction Motor Dataset from Kaggle. An artificial dataset was created with the same statistical properties, where six conditions were used, such as the Normal, Broken Rotor Bar, Stator Fault, Bearing Fault, Voltage Unbalance, and Overload. There are 128 time points in each sample and 32 engineered features. Table I indicates the distribution of the dataset.

TABLE I
Dataset Class Distribution

Class	Samples	Percentage
Normal	1666	16.67%
Broken Rotor Bar	1666	16.67%
Stator Fault	1666	16.67%
Bearing Fault	1666	16.67%
Voltage Unbalance	1666	16.67%
Overload	1666	16.67%
Total	9996	100%

B. Experimental Setup

Intel Core i7-10750H with 16GB RAM was experimented on. Split of data: training (6997), validation (1499) and test (1500) in stratified sampling.

Training hyperparameters: 30 epochs, batch size 32, Adam optimizer with initial learning rate 0.001, categorical cross-entropy loss.

C. Evaluation Metrics

Performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.

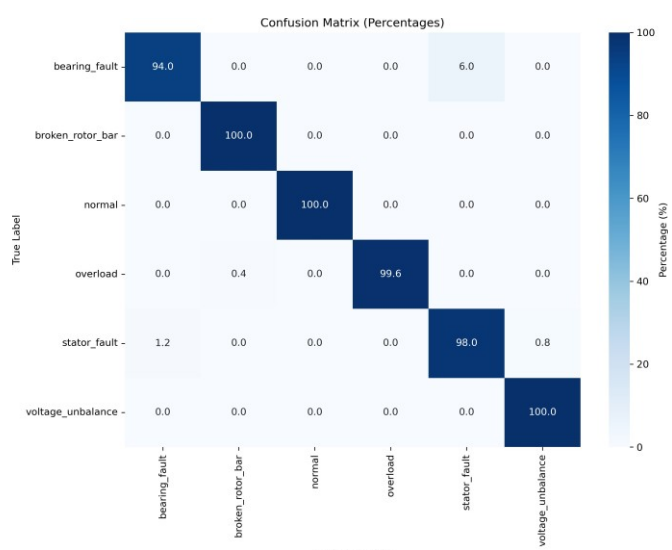

Fig. 2. Confusion matrix visualization showing classification performance across all six fault conditions

V. Results and Discussion

A. Overall Performance

The presented model scored extraordinarily with regard to the test set. Table II is a summary of overall metrics.

TABLE II
Overall Model Performance

Metric	Score
Test Accuracy	0.9860
Test Precision (weighted)	0.9863
Test Recall (weighted)	0.9860
Test F1-Score (weighted)	0.9860
Training Accuracy (final)	0.9907
Validation Accuracy (final)	0.9887

B. Per-Class Analysis

Per-class breakdown is given in Table III. The model has scores of 1.00 on broken rotor bar, normal and overload classes. There is minor confusion in bearing and stator faults.

TABLE III
Per-Class Classification Report

Class	Precision	Recall	F1-Score	Support
bearing_fault	0.99	0.94	0.96	250
broken_rotor_bar	1.00	1.00	1.00	250
normal	1.00	1.00	1.00	250
overload	1.00	1.00	1.00	250
stator_fault	0.94	0.98	0.96	250
voltage_unbalance	0.99	1.00	1.00	250
Macro Avg	0.99	0.99	0.99	1500

C. ROC-AUC Analysis

Table IV shows ROC-AUC scores. The classes of AUC are above 0.99 with four classes having a 1.00. Micro-average AUC is 0.9998.

TABLE IV
ROC-AUC Scores by Class

Class	AUC Score
bearing_fault	0.9979
broken_rotor_bar	1.0000
normal	1.0000
overload	1.0000
stator_fault	0.9977
voltage_unbalance	1.0000
Micro-average	0.9998

D. Training Dynamics

This model achieved high validation accuracy of over 95% in the first two epochs. Training and validation curves are close, which implies that there is not much overfitting, owing to considerable regularization. The best model was saved in epoch 27 and the validation accuracy was 98.93%.

E. Comparative Analysis

SVM using RBF kernel reached the accuracy of 92.5% based on the identical set of features. Standalone CNN (96.2%) standalone LSTM (95.8%): there is meaningful improvement in the hybrid approach (98.6%). Our attention-enhanced architecture is an improvement of the one by Shao et al. [9] (97.5%) by 1.1% in absolute terms.

F. Sample Predictions

Table V shows sample predictions with high confidence scores.

TABLE V
Sample Predictions

True Class	Predicted Class	Confidence
stator_fault	stator_fault	0.998
overload	overload	1.000
overload	overload	1.000

G. Discussion

It has been observed that exceptional performance can be attributed to: (1) multi-scale CNNs to the capture of fault signatures in various resolutions, (2) stacked BiLSTMs to create more abstract representations of time, (3) to attention mechanism, which concentrates on diagnostically relevant parts, and (4) to comprehensive regularization to eliminate overfitting.

The combination of engineered features with deep learning features produces integrated information—engineered features extract domain-specific knowledge and deep learning features automatically learn latent patterns, resulting in a more stable feature representation.

One limitation is that bearing and stator faults can produce similar spectral signatures, introducing some uncertainty and driving further feature engineering.

H. Conclusion and Future Work

This paper presented an AI-based predictive maintenance system for induction motors using a hybrid CNN-LSTM-Attention framework. The proposed model captures multi-scale spatial features through parallel CNN branches with kernel sizes of 3, 5, and 7, while the stacked BiLSTM network models temporal dependencies. The attention mechanism selectively focuses on diagnostically relevant time steps, enhancing both interpretability and classification accuracy. Experimental evaluation achieved a test accuracy of 98.60

Future work includes validation on real-world industrial datasets, implementation of transfer learning for cross-domain adaptation, development of explainable AI techniques to enhance interpretability, and optimization for edge deployment to enable real-time monitoring in Industry 4.0 environments. Additionally, we plan to extend the framework for remaining useful life (RUL) estimation and incorporate multi-sensor fusion to further improve predictive maintenance abilities.

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